

Individual Assignment 5

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Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c(" ", "n/a", "N/A", "over", "173+")
)

```

Cleaning Data

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

```

```

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join to only include sites in ncos
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with data frames

```

```

unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

```

```

aq_ins_clean <- aq_ins |>
clean_names() |>
semi_join(ncos_sites, by = "site") |>
rename (date = date_on_vial) |>
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
filter(sample_type %in% c("FB250", "FB 250")) |>
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>
group_by(date, site, taxon) |>
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list

```

```

left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
))

```

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "red",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "Median salinity (ppt)",
  title = "Site salinity") +
# different theme
theme_bw()

```

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

```

```

# base layer
copepods <- ggplot(data = copepods,
                  # x-axis: site
                  # y-axis: mean abundance/L
                  mapping = aes(x = site_full,
                               y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a)

- x-axis: site
- y-axis: log + 1 transformed average abundance/L
- geom to represent average abundance/L: `geom_jitter()`
- geom to represent medians: `stat_summary()`

b)

```

# creating new object from clean aquatic invertebrates
ostracod <- aq_ins_clean |>
# filter to only include ostracod
filter(taxon == "ostracod") |>

```

```
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

ostracod |>
  slice_sample(n = 10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2021-08-25_NPB	2021-08-25 00:00:00	NPB	ostr~	581.	7.52	4.57	4.5
2	2022-08-12_NVBR	2022-08-12 00:00:00	NVBR	ostr~	NA	NA	NA	NA
3	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.3	NA	NA	NA
4	2023-05-16_NVBR	2023-05-16 00:00:00	NVBR	ostr~	NA	9.29	NA	9.83
5	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	ostr~	NA	8.86	45.2	10.2
6	2022-08-10_NPB	2022-08-10 00:00:00	NPB	ostr~	2.8	8.28	8.84	9.81
7	2023-05-05_NPB	2023-05-05 00:00:00	NPB	ostr~	24.8	NA	NA	12.1
8	2022-11-19_NDC	2022-11-19 00:00:00	NDC	ostr~	6.13	NA	NA	NA
9	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.4	NA	NA	NA
10	2022-05-06_NMC	2022-05-06 00:00:00	NMC	ostr~	NA	9.96	1.45	16.5

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <chr>, log_ave_lit <dbl>

c)

```
# base layer
ostracod_plot <- ggplot(data = ostracod,
  mapping = aes(x = site_full,
    y = log_ave_lit)) +

  # first layer: points to represent observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +

  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    color = "pink") +

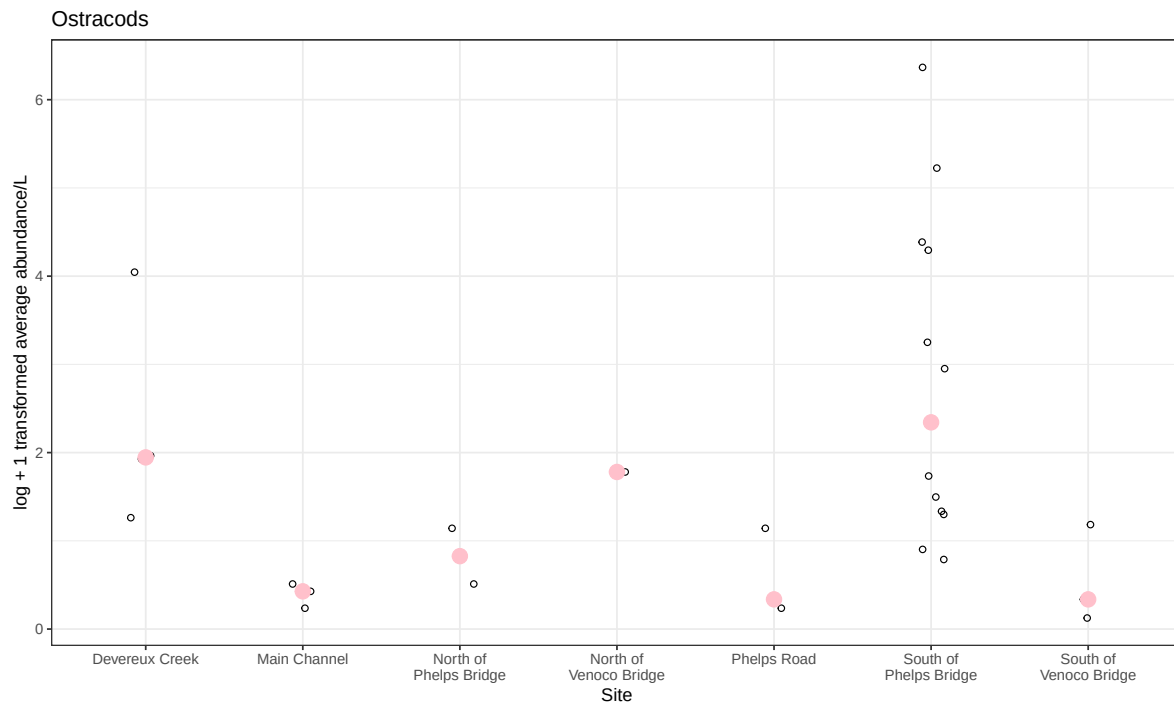
  # wrapping x-axis text
```

```

scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Ostracods") +
# different theme
theme_bw()

# displaying object
ostracod_plot

```

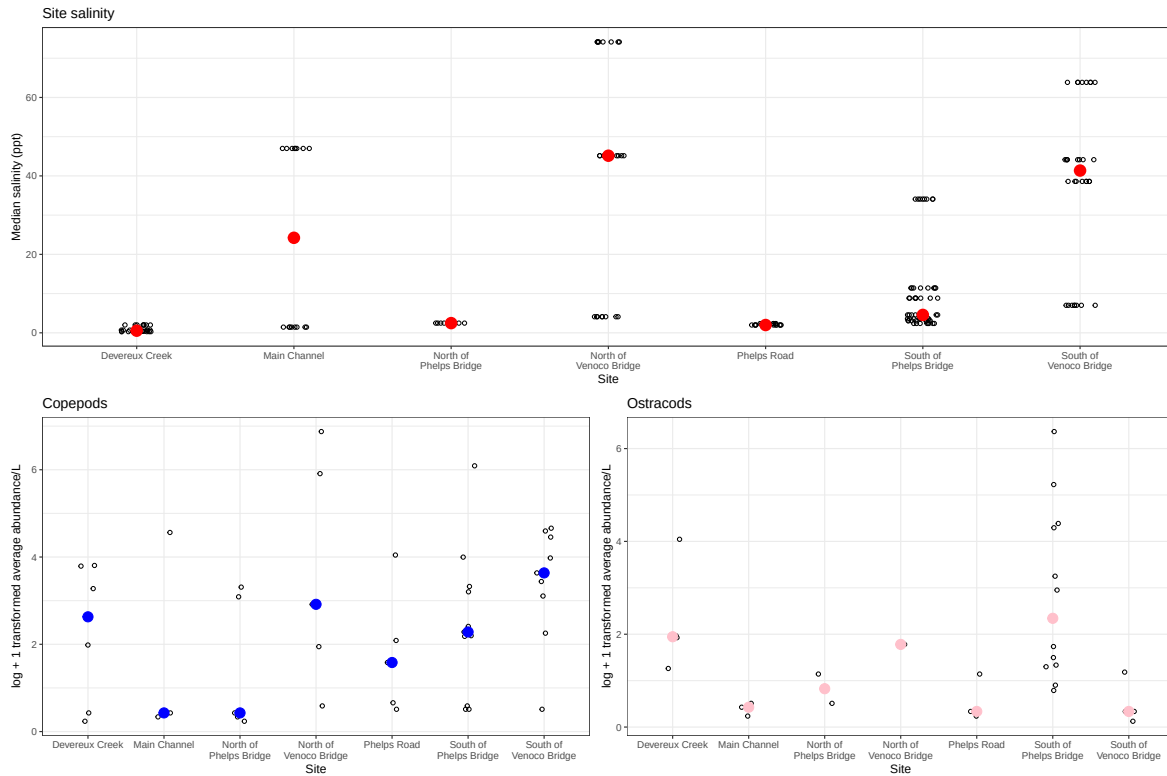


d)

```

# create multipanel figure
salinity / (copepods + ostracod_plot)

```



e)

Copepods are more abundant than ostracods at these sites: Devreux Creek, North of Venoco Bridge, Phelps Road, and South of Venoco Bridge.

It does not appear that ostracod abundance relates to site-level differences in salinity. At South of Venoco Bridge salinity is high while abundance is low however at North of Venoco Bridge both salinity and abundance are high.

2. Visualizing total abundance as a function of salinity

a)

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_jitter()
- facets: facet_wrap(~ taxon, scales = "free")

b)

```
# creating new object from clean aquatic inverts
aq_ins_salinity <- aq_ins_clean |>
  # create log + 1 transformed average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1),
    # make taxon names title case
    taxon = str_to_title(taxon)) |>
  # order taxon levels by maximum abundance
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep columns needed for figure
  select(taxon, ave_lit, log_ave_lit, med_sal, site_full)

# display 10 random rows
aq_ins_salinity |>
  slice_sample(n = 10)
```

A tibble: 10 x 5

	taxon	ave_lit	log_ave_lit	med_sal	site_full
	<fct>	<dbl>	<dbl>	<dbl>	<chr>
1	Ostracod	NA	NA	NA	Phelps Road
2	Diptera_ceratopogonidae	NA	NA	4.1	North of Venoco Bridge
3	Annelida_polychaete	NA	NA	11.4	South of Phelps Bridge
4	Annelida_polychaete	6	1.95	NA	South of Phelps Bridge
5	Ostracod	0.4	0.336	63.8	South of Venoco Bridge
6	Annelida_polychaete	NA	NA	NA	Main Channel
7	Copepod	36.9	3.64	NA	South of Venoco Bridge
8	Ostracod	NA	NA	NA	South of Venoco Bridge
9	Diptera_ceratopogonidae	NA	NA	63.8	South of Venoco Bridge
10	Annelida_polychaete	NA	NA	NA	Phelps Road

c)

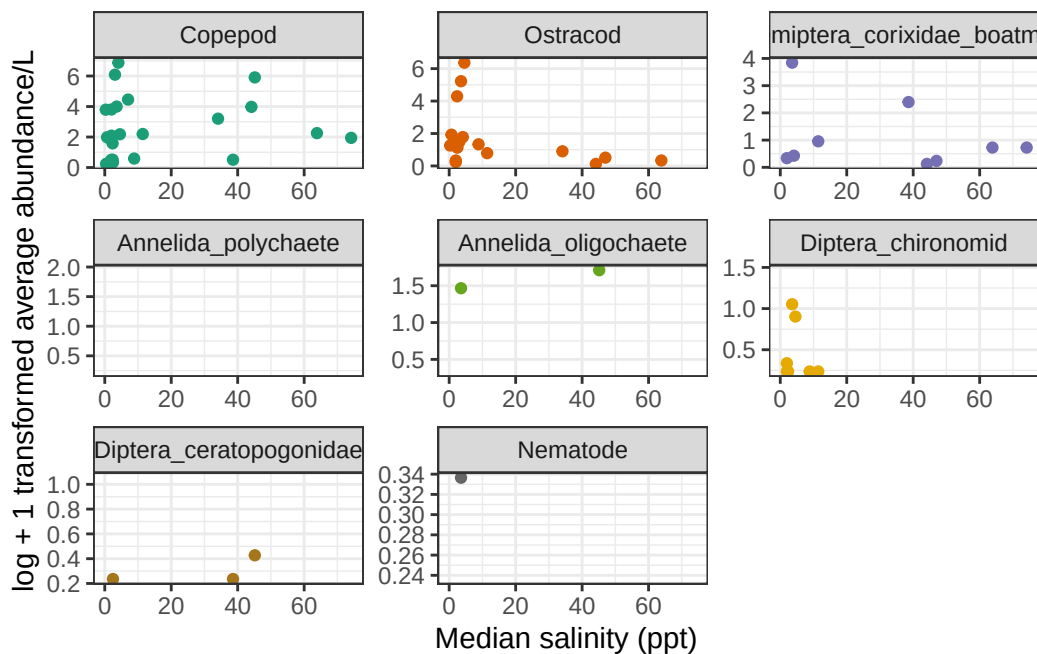
```
salinity_taxa <- ggplot(data = aq_ins_salinity,
  mapping = aes(x = med_sal,
    y = log_ave_lit,
    color = taxon)) +
  geom_point() +
  # separate panel for each taxon
```

```

facet_wrap(~ taxon, scales = "free") +
# custom colors
scale_color_brewer(palette = "Dark2") +
# labels
labs(x = "Median salinity (ppt)",
      y = "log + 1 transformed average abundance/L") +
# remove legend
theme_bw() +
theme(legend.position = "none")

# display figure
salinity_taxa

```



d)

For Annelida Polychaete, Annelida Oligochaete, and Nematode, not much can be said for the relationship between abundance and salinity due to a lack of data. For Copepod and Ostracods it seems that both species are the most abundant at low mean salinities. It also seems that Copepods are more resilient to higher salinities because their abundance remains high even past 60 ppt while Ostracod abundance decreases. Diptera Chironomid are only abundant at low salinities and Boatmen have similar abundances across all salinities.